

Global Views on AI: Assignment 8

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```
library(ggplot2)
library(dplyr)
library(tidyr)
```

Introduction

In this report, I visualize the global public perception of artificial intelligence based on the Pew Research Center's Spring 2025 survey. I analyze three key dimensions: Awareness, Sentiment, and Trust in Regulation.

```
# 1. Awareness Data
veri_awareness_tablo <- "Country,A_lot_pct,A_little_pct,Nothing_at_all_pct
U.S.,47,48,5
Canada,41,51,8
France,52,40,8
Germany,51,45,4
Sweden,46,45,8
Netherlands,46,44,10
Italy,45,46,9
UK,41,49,10
Hungary,38,53,8
Poland,34,53,13
Greece,34,49,17
Spain,30,56,14
Japan,53,36,11
Australia,44,53,3
South Korea,21,57,21
Indonesia,18,36,43
India,14,32,35
Israel,36,44,18
Turkey,19,50,25
South Africa,30,31,34
Nigeria,17,44,32
Kenya,12,36,49
Argentina,24,48,28
Brazil,22,47,30
Mexico,19,53,27
25-country median,34,47,14"

df_awareness <- read.csv(text = veri_awareness_tablo)

# 2. Sentiment Data
```

```

veri_sentiment_tablo <- "Country,More_concerned_than_excited_pct,Equally_concerned_and_excited_pct,More
U.S.,50,38,10
Italy,50,37,12
Australia,49,38,13
Brazil,48,37,10
Greece,47,39,10
Canada,45,45,9
UK,39,46,13
Argentina,39,41,13
Spain,39,38,19
Poland,37,42,15
Mexico,35,47,13
France,35,49,15
Netherlands,34,48,16
Hungary,33,47,18
Indonesia,32,49,14
Kenya,31,43,17
Sweden,31,45,22
South Africa,30,42,18
Germany,29,53,17
Japan,28,55,16
Turkey,26,35,19
Nigeria,24,36,20
Israel,21,34,29
India,19,39,16
South Korea,16,61,22
25-country median,34,42,16"

```

```

df_sentiment <- read.csv(text = veri_sentiment_tablo)

```

3. Trust Data

```

veri_trust_tablo <- "Country,Trust_Own_Country_pct,Trust_EU_pct,Trust_US_pct,Trust_China_pct
U.S.,44,43,44,13
Canada,64,57,33,17
France,47,47,21,17
Germany,70,71,33,23
Greece,22,38,37,34
Hungary,56,56,56,43
Italy,37,42,32,33
Netherlands,68,68,35,25
Poland,53,44,37,13
Spain,55,61,34,31
Sweden,55,54,25,15
UK,57,56,37,24
Australia,65,59,30,15
India,89,44,64,27
Indonesia,74,58,54,64
Japan,41,43,41,17
South Korea,55,53,58,21
Israel,72,54,70,22
Turkey,60,36,23,32
Kenya,54,58,62,61
Nigeria,46,72,79,79

```

```

South Africa,64,42,50,57
Argentina,33,31,35,39
Brazil,36,26,35,32
Mexico,50,35,24,38
25-country median,55,53,37,27"

```

```
df_trust <- read.csv(text = veri_trust_tablo)
```

AI Awareness Levels by Country

To understand how much people know about AI, I created an ordered stacked bar chart. This allows us to compare the level of familiarity across different nations.

```

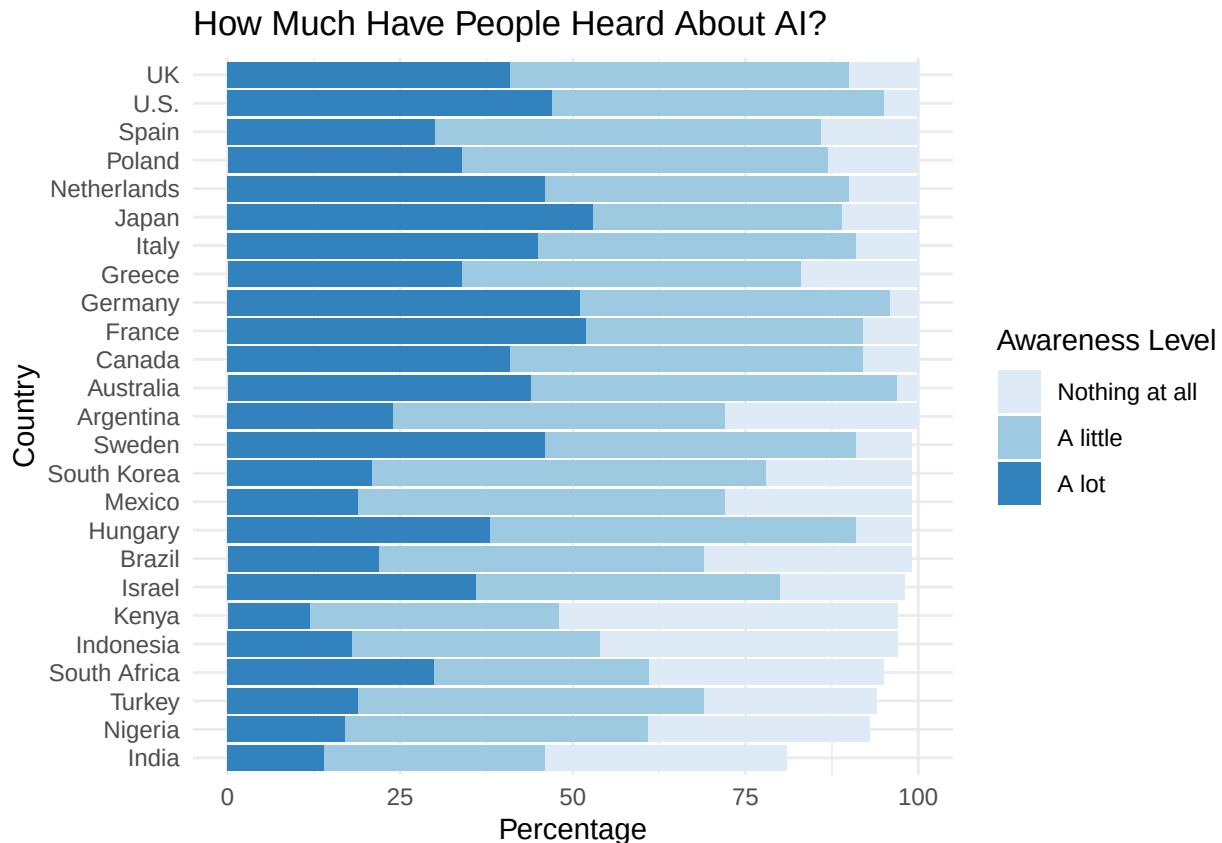
# We convert data to long format
df_awareness_long <- df_awareness %>%
  filter(Country != "25-country median") %>% #I'm removing the median line so it doesn't appear as a separate country
  pivot_longer(
    cols = c("A_lot_pct", "A_little_pct", "Nothing_at_all_pct"),
    names_to = "Awareness_Level",
    values_to = "Percentage"
  )

#Let's arrange the factor levels (so that they are sorted properly on the chart)
df_awareness_long$Awareness_Level <- factor(
  df_awareness_long$Awareness_Level,
  levels = c("Nothing_at_all_pct", "A_little_pct", "A_lot_pct"),
  labels = c("Nothing at all", "A little", "A lot")
)

# Let's rank the countries according to the 'A lot' rate
df_awareness_long$Country <- reorder(df_awareness_long$Country, df_awareness_long$Percentage, FUN = sum)

# graphic sketching
ggplot(df_awareness_long, aes(x = Country, y = Percentage, fill = Awareness_Level)) +
  geom_bar(stat = "identity") +
  coord_flip() + # For horizontal bar chart
  labs(
    title = "How Much Have People Heard About AI?",
    x = "Country",
    y = "Percentage",
    fill = "Awareness Level"
  ) +
  theme_minimal() +
  scale_fill_brewer(palette = "Blues") # we use blue

```



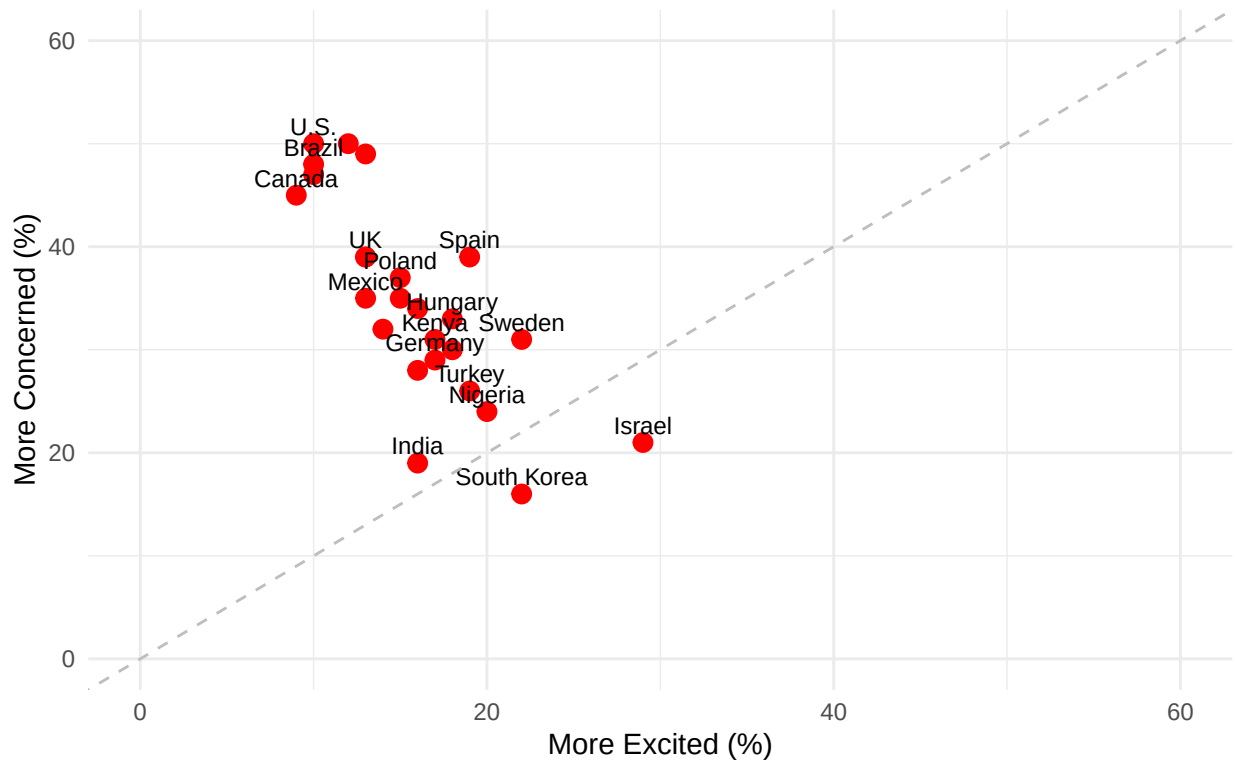
Findings: The data reveals a significant “awareness gap” between developed and developing nations. Countries with advanced technological infrastructures and high GDP per capita, such as Japan (53% heard ‘a lot’), the United States (47%), and Germany, exhibit high levels of familiarity with AI. In stark contrast, emerging economies like Kenya, India, and Indonesia have substantial populations who report hearing “nothing at all” about AI (reaching up to 49% in Kenya). This disparity suggests that the global AI discourse is still primarily concentrated in wealthier, more digitized societies, while a large portion of the Global South remains on the periphery of these discussions.

##C oncern vs. Excitement (Sentiment Analysis) This scatter plot explores the balance between concern and excitement regarding AI’s increasing use.

```
df_sentiment_plot <- df_sentiment %>%
  filter(Country != "25-country median")

ggplot(df_sentiment_plot, aes(x = More_excited_than_concerned_pct, y = More_concerned_than_excited_pct)) +
  geom_point(color = "red", size = 3) +
  geom_text(aes(label = Country), vjust = -0.5, size = 3, check_overlap = TRUE) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray") + # x=y line
  labs(
    title = "AI Sentiment: Concern vs. Excitement",
    x = "More Excited (%)",
    y = "More Concerned (%)",
    caption = "Dashed line represents equal concern and excitement"
  ) +
  theme_minimal() +
  xlim(0, 60) + ylim(0, 60)
```

AI Sentiment: Concern vs. Excitement



Dashed line represents equal concern and excitement

Findings: The scatter plot illustrates a distinct geographical and cultural divide in AI sentiment. Most Western nations, particularly the United States, Australia, and Western European countries, are positioned significantly above the dashed line ($y = x$), indicating that skepticism prevails; their populations are far more “concerned” than “excited.” Conversely, countries in the Global South (such as India, Nigeria, and Kenya) and some Asian nations tend to cluster closer to the line or below it, showing relatively higher excitement or a balanced view. This implies that while developed nations—where AI disruption is imminent—are wary of risks like job loss and privacy, developing nations may perceive AI more optimistically as a tool for economic development and opportunity.

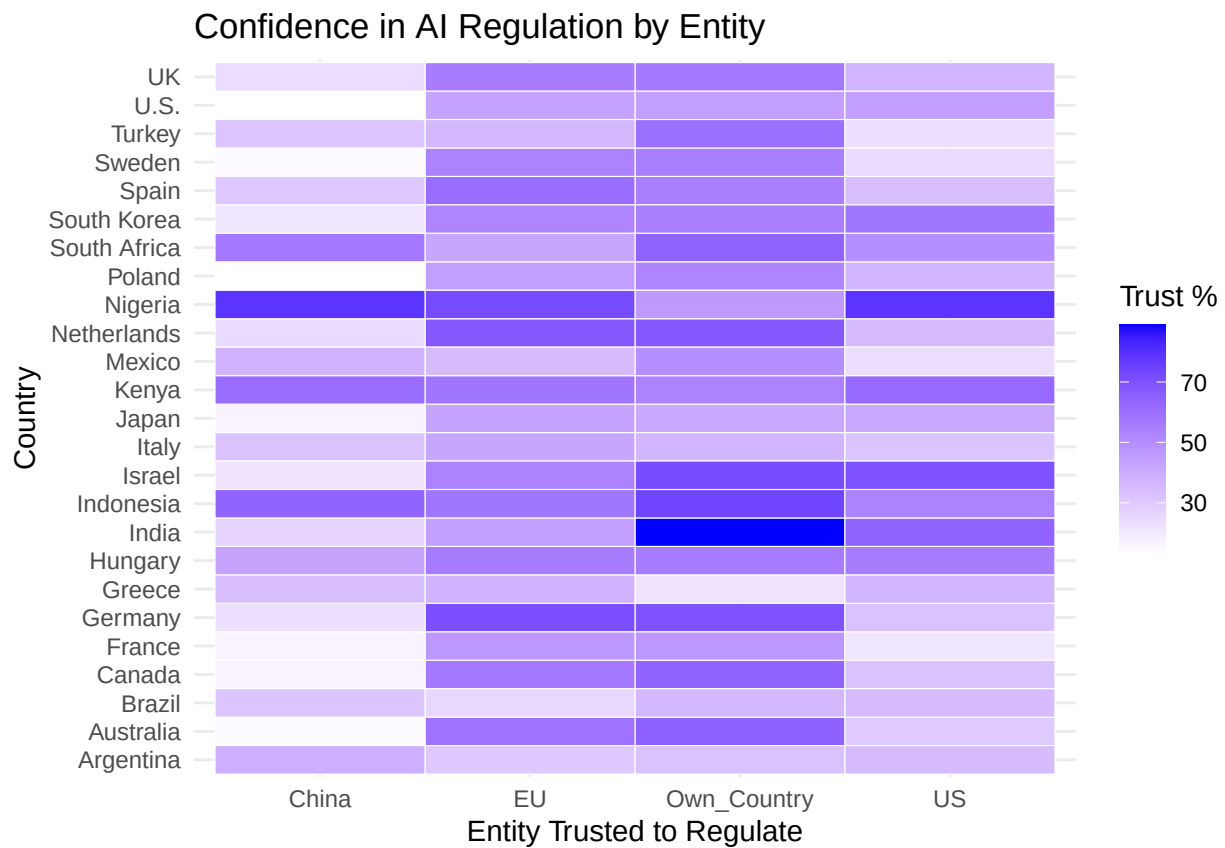
Trust in AI Regulation

Finally, I use a heatmap to visualize which entities people trust to regulate AI effectively

```
# We convert data to long format again
df_trust_long <- df_trust %>%
  filter(Country != "25-country median") %>%
  pivot_longer(
    cols = c("Trust_Own_Country_pct", "Trust_EU_pct", "Trust_US_pct", "Trust_China_pct"),
    names_to = "Entity",
    values_to = "Trust_Score"
  )

# I clean up the names (removing the underscores)
df_trust_long$Entity <- gsub("Trust_", "", df_trust_long$Entity)
df_trust_long$Entity <- gsub("_pct", "", df_trust_long$Entity)
```

```
ggplot(df_trust_long, aes(x = Entity, y = Country, fill = Trust_Score)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "white", high = "blue") + # Dark color = High trust
  labs(
    title = "Confidence in AI Regulation by Entity",
    x = "Entity Trusted to Regulate",
    y = "Country",
    fill = "Trust %"
  ) +
  theme_minimal()
```



Findings: The heatmap highlights a geopolitical fragmentation in trust regarding AI governance. The most prominent trend is local bias: populations generally display the highest confidence in their own national governments or immediate regional blocs (e.g. European nations showing high trust in the EU).

Furthermore, a polarization is visible regarding global superpowers. While there is a moderate to high trust in the U.S. from its traditional allies (e.g. Israel, South Korea), trust in China as a regulator is exceptionally low across most Western democracies. Interestingly, African and Southeast Asian nations like Kenya, Nigeria, and Indonesia display a more multi-aligned approach, showing comparatively higher trust in both Chinese and American regulations than their Western counterparts.